

Supplementary material for CONTAINER: Bounding Credential-Compromise Impact in IEEE 2030.5 DER Systems

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S1. The IEEE 9500 non-convergence, in full

The manuscript’s limitations paragraph states that the PNNL 9500-node system did not converge in our build and therefore contributes no reported number. This section names the failure mode, so that the absence is an explained negative result rather than a silent omission.

The failure is in the *base case*, with no attack and no PV of ours placed: the model’s own shipped driver, unaltered, returns `Converged = False` with an empty error description and every bus per-unit magnitude reading zero — the solution collapses rather than missing tolerance. It reproduces at 1000 iterations, under both the normal and Newton algorithms, and with feeder controls enabled and disabled, so it is neither a control-loop oscillation nor an iteration-budget shortfall. Our build is DSS C-API 0.14.5 / DSS-Python 0.15.7 / OpenDSSDirect.py 0.9.4 on macOS arm64.

We did not resolve it and we do not attribute it, having had no second OpenDSS build to discriminate a defect on our side from one in the model. The consequence for the manuscript is stated there and repeated here because it is load-bearing: cross-feeder generality rests on nothing we report, and every physical number in the paper comes from the single IEEE 8500-node feeder.

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S2. Seed inventory for the pilot evaluation

Table 1 inventories every measured run behind the manuscript’s pilot evaluation, so no experiment’s scale has to be inferred from prose. Seeds are paired across arms within an experiment; “arms” counts the policy or envelope conditions evaluated at each seed. The runs made on the superseded harness — withdrawn in the manuscript and not reported as results — are inventoried separately in `results/superseded/`.

Table 1: Inventory of the measured runs the pilot evaluation reports.

Experiment	Seeds	Arms	States
IEEE 8500 time series	20	9	2
Credential-lifetime sweep	20	18	2
Scope-envelope sweep	5	21	2
Volt-var-curve sweep	5	18	2
Control-mode diagnostic	1	2	1
Credential-service invariants (C1)	1	10	—
Overhead microbenchmarks (C3)	500 iterations	4	—
Availability fault injection (C6)	3 policies	3	—